Media Wall & Dining Wall: Build Guide

Version 11 · Oct 18, 2026 · for Emma and Dave · sizes in cm; the full plan has the drawings and the fit proof



This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Where a step is a job for a professional, it says so. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **Mains wiring is the electrician's.** Swapping the socket, the new socket behind the sofa and the new ceiling point are all theirs. Adding to existing circuits like this isn't normally notifiable under Part P, but a registered electrician certifies it either way.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15

Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.

2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	**Measure**	The alcove's width at six heights, the floor's level, the breast's flatness, the door's swing, the sofa's size; count your records and games.	Numbers within 1 cm of the plan, or tell me and I'll redraw.	2 h
2	**Network, stage 1**	Hub in modem mode; Decos and switches over Wi-Fi.	Internet in every room.	2 h
3	**Electrician**	Cat6 runs, the double socket, the socket behind the sofa, the second ceiling point and dimmer, the fused spur. Mark every cable route on the wall.	The certificate, and photos of where the cables run.	1–2 days (theirs)
4	**Paint**	The whole room in Card Room Green; skirting and door in eggshell. Let it cure a week before fixing anything.	Two even coats; no white showing at the edges.	2–3 days
5	**Mark the heights**	With the laser: 73 and 76 (worktop), 130 and 134 (Series X shelf), 153 and 155.4 (ledges), 192 and 196 (high shelf); the uprights at 0–4.5 and 131–135.5 from the side wall; the dining wall's backboard at 51.3–73.3, centred on the breast.	Lines level end to end; the uprights' lines plumb.	1 h
6	**Base**	Build the plinth (with the drawer bay under bay 2), level it with shims, stand the two BESTÅ frames on it; build bay 3 from ply and bolt it on. Spacers to the wall, anti-tip brackets, trunking in the channel, cables run in.	The tops level within 1 mm across 186; nothing rocks.	1 day
7	**Worktop**	Scribe the worktop to the walls, cut the TV slot and the grommet hole, oil all over, screw it up through the tops in slotted holes.	Tight to the walls; the slot over the channel.	½ day
8	**Inside the base**	Shelves at the fit-proof heights; the game drawer; the record trays' runners at 2.3 and 37.0, then the trays; the plinth drawer.	Every drawer runs smoothly; trays clear each other by 2 cm.	1–2 days
9	**Uprights and shelves**	Cut the housings and mortises on the bench. Dry-fit each upright with its shelves. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the brackets, then fit the high shelf, the ledge, the Series X shelf (wedge the tenons) and the pegs.	Uprights plumb; shelves level; the TV slides between with 2 cm each side.	2–3 days

10	**Side wall**	The oak cleat for the wrap ledge and the PC shelf; dowel the wrap ledge to the left upright's front edge; the concealed PC brackets; the cane screen; the trunking down to the skirting.	One level line with the ledge over the TV; the PC shelf takes your weight leaning on it.	1 day
11	**Dining wall**	Screw the backboard to the breast (plugs every 20 cm). Fit the two Hebgo brackets on its face, 80 apart, arm tops level with its top. Screw the lift-off hinges to its top edge and the table's back edge. Hang the table, fit the turn buttons at the top, fit the leg plates underneath.	Folds up and down without binding; brackets lock both sides; it doesn't move when you lean on it.	1 day
12	**Lighting**	LED in the upright grooves, the ceiling strip through its notches, the bias light; aim both spot bars.	No head shines at the screen or into your eyes from the sofa.	½ day
13	**Connect**	TV, soundbar, consoles, deck, PC; network stage 2; label every plug and input.	The "done when" list in the network section.	1 day

Roughly 12–15 working days spread over a few weekends, plus the electrician and the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the trays before the anti-tip brackets go in.** A full tray weighs about 32 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

Shopping list

About £2,527–4,757 plus the electrician, in build order.

Status	Item	Qty	Estimate (£)	Paid (£)
1 NETWORK, NOW				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
2 ELECTRICIAN AND CABLING				
☐ To do	Book one electrician visit: Cat6 from bedroom 3 down through the airing cupboard to the alcove's right corner (Deco B) and on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2; swap the side-wall socket for a double **at the same spot**; a new double socket low on the side wall behind the sofa for the PC and the rear speakers **(new)**; a new ceiling point by the stairs for the second spot bar, with a two-gang LED dimmer by the door **(new)**; a fused spur for the LED drivers; fit both spot bars		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To buy	Slim paintable trunking: the PC drop from the shelf to the skirting and along it to the base, about 4 m **(changed: longer)**	1	20–35	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 lead, shelf-notch grommet	1	10–20	
3 PAINT				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door, frame and trunking; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
4 BASE UNIT				
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☐ To do	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid, or plasterboard on studs **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6, 12 mm ply box 52 × 34 × 5, 300 mm push-to-open runners **(moved from bay 3)**	1	30–50	
☐ To buy	Bay 3 carcass for the records: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts **(changed: holds the trays)**	1	45–80	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To buy	Compact isolation riser for the sub, about 30 × 35 cm, 4–6 cm tall	1	20–40	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
5 OAK JOINERY				
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, wrap ledge 134 × 7 × 2.4, PC shelf 82 × 28 × 4, lips **(changed)**	1 set	200–340	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Cane screen for the PC: oak frame and pre-woven cane, 26 × 48 cm	1	20–40	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
6 DINING WALL				
☐ To buy	Solid oak table top 110 × 80 × 2.7 and a 60 × 80 × 2.7 leaf (edge-glued staves from a timber merchant, or cut from a 90 cm deep oak worktop) **(new)**	1 set	200–380	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Eight Emuca Ø60 × 710 mm screw-in table legs with plates (two sets of four): four for the top, four for the leaf **(new)**	2 sets	70–140	
☐ To buy	Two oak turn buttons, two leaf-alignment dowels and two table clips for the leaf, felt pads **(new)**	1	10–20	
☐ To buy	Four folding chairs, IKEA TERJE or similar (skip if yours fit: seat about 45–48) **(new)**	4	100–160	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Fix the backboard to the breast, hang the brackets and the top, check it folds and locks; set the legs to the right height			
7 LIGHTING				
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	1	40–90	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars (£49 each), plus eight warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed: two)**	2	120–160	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
8 TV, CONSOLES AND AV				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
9 VINYL KIT				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
10 FINISHING				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			